PUNE VIDYARTHI GRIHA's
COLLEGE OF ENGINEERING AND TECHNOLOGY
&
G K PATE (WANI) INSTITUTE OF MANAGEMENT, PUNE – 411 009

INTERNSHIP REPORT

ON

OTP based smart locking system

Under Supervision of

Mr. Manoj Gunjikar, Team leader

Veblogy innovative technology pvt. Ltd.

(Date – 23/12/2024 to 23/1/2025)

SUBMITTED BY
Neha Kanhu Katake (2172)

Class TE (E&TC)
April-May 2025

DEPARTMENT OF E&TC ENGINEERING

PVG's COET & GKPIOM, PUNE-9

SAVITRIBAI PHULE PUNE UNIVERSITY

DEPARTMENT OF E&TC ENGINEERING

PUNE VIDYARTHI GRIHA's
COLLEGE OF ENGINEERING AND TECHNOLOGY
&
G K PATE (WANI) INSTITUTE OF MANAGEMENT, PUNE – 411 009

CERTIFICATE

This is to certify that the "Internship report" submitted by **Neha kanhu katake(T1900703075),** is work done by her and submitted during 2024 - 2025 academic year, in partial fulfillment of the requirements for the Third Year Course in Electrical Engineering, Pune Vidyarthi Griha's College Of Engineering & Technology And G K Pate (Wani) Institute of Management, Pune – 411 009

Prof. N. D. Kuchekar, Prof. K. C. Wanzare
Department Internship Coordinator

Dr. P. G. Shete
Head of the Department
Department of E&TC Engineering

Certificate of Internship

VEBLOGY
INNOVATIVE TECHNOLOGY PVT LTD

Date: 24th January 2025

Internship Completion Certificate

This is to certify that **Ms. Neha Kanhu Katake,** a student of Pune Vidhyarthi Grihas College Of Engineering And Technology, has successfully completed his internship project at **Veblogy Innovative Technology Pvt Ltd.** from **23rd December 2024 To 23rd January 2025.**

During the internship with us, she was found punctual, hardworking and inquisitive.

She was working under a senior team member in development of the project.

Project Name: OTP Based Smart Wireless Locking System Using Arduino

Technology: Arduino Microcontroller, Wireless Communication (Wifi, Bluetooth)

We wish her every success in life.

VEBLOGY INNOVATIVE TECHNO... PUNE

Project Guide

Vishal Chinchane
(Project Manager)

B206, Specialty Business Center, Balewadi, Pune-411045. Phone: +91-020-64102149 email: info@veblogy.com

Department Certificate

Pune Vidyarthi Griha's

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Formerly known as PVG's COLLEGE OF ENGINEERING AND TECHNOLOGY, PUNE

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TO,

Date:

Dear Sir/Madam,

PVG's College of Engineering & Technology and G. K. Pate (Wani) Institute of Management is well known for its Printing Engineering branch since 1985. The other branches of Mechanical, Electrical, and Electronics & Telecommunication Engineering were added in 1991. The college has also started the branches of Information Technology, Computer Engineering in 2001 and 2002 respectively and recently Artificial Intelligence & Data Science branch is introduced from 2021. The college is run by PVG COET & GKPIOM, Pune, which is a well-known educational and charitable institute in Maharashtra.

Recognizing the importance of practical industry exposure, we encourage our students to undergo internships and projects within reputed organizations. Therefore, I am pleased to recommend number of students from our college, who are currently pursuing studies in Second Year (SE), Third Year (TE), and Fourth Year (BE) to participate in an internship at your esteemed organization for a duration of ______number of weeks/months. The names of the recommended student/s are as follows:

1.
2.

It is important to note that these students will-make their own arrangements and will not pose any financial burden on your organizatiori. Furthermore, they will strictly adhere to your company's rules and regulations throughout their internship period. We kindly request that, upon the completion of their internship, your organization provides feedback on their performance on official letterhead, as its requirement guided by SPPU for evaluation of student duly signed by an authorized representative.

Your collaboration would be a valuable opportunity for our students to gain practical experience and enhance their industry knowledge. We anticipate a positive response from your end.

We look forward to a mutually beneficial partnership.
Thanking you,
Yours sincerely,

Prof. N. D. Kuchekar
Internship Coordinator
Dept of E & TC
Email id: ndk_entc@pvgcoet.ac.in
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Dr. P. G. Shete
H. O. D
Dept of E & TC

Dr. M. R. Tarambale
Principal
PVG's COET & GKPIOM, Pune

PVG's COET & GKP(W) IOM PUNE-09

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ACKNOWLEDGEMENT

First I would like to thank Mr *Mr. Manoj Gunjikar, Team leader Veblogy innovative technology pvt. Ltd.* for giving me the opportunity to do an internship within the organization.

I also would like all the people that worked along with me in *Veblogy innovative technology pvt. Ltd.* with their patience and openness they created an enjoyable working environment.

It is indeed with a great sense of pleasure and immense sense of gratitude that I acknowledge the help of these individuals.

I am highly indebted to Principal Dr. M. R. Tarambale, for the facilities provided to accomplish this internship.

I would like to thank my Head of the Department Dr. P. G. Shete for his constructive criticism throughout my internship.

I would like to thank Prof. N. D. Kuchekar, Prof. K. C. Wanzare Department internship coordinator for his support and advices to get and complete internship in above said organization.

I am extremely great full to my department staff members and friends who helped me in successful completion of this internship.

Neha Kanhu Katake(T1900703075)

ABSTRACT

The OTP-Based Smart Wireless Locking System using Arduino is an innovative solution designed to enhance the security and convenience of traditional locking mechanisms. This system leverages the capabilities of Arduino microcontroller and wireless communication technologies to create a secure and user-friendly lock system that operates based on One-Time Password (OTP) authentication. To unlock the system, users need to generate an OTP using a designated mobile application. The OTP is securely transmitted to the central control unit via a wireless communication protocol such as Bluetooth. Upon verification of the OTP's validity, the central control unit sends the corresponding command to the designated wireless lock module, instructing it to unlock. The system incorporates several advanced features to enhance security and functionality. Firstly, the OTP ensures a higher level of security by utilizing unique passwords for every authentication attempt, minimizing the risk of unauthorized access. The OTP-Based Smart Wireless Locking System using Arduino offers a reliable, secure, and user-friendly solution for modern access control needs

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Worksheet of Internship

{For Each Week}

	DATE	DAY	NAME OF THE TOPIC / MODULE COMPLETED
1st WEEK	23-12-25	MONDAY	Interaction with industry people and team leader
	24-12-25	TUESDAY	Orientation about company , their policies, work ethics and other guidelines
	25-12-25	WEDNESDAY	Christmas
	26-12-25	THURSDAY	Introductory session about IoT , explored internship objectives and expectations
	27-12-25	FRIDAY	Overview of different IoT components , architecture etc

	DATE	DAY	NAME OF THE TOPIC / MODULE COMPLETED
2st WEEK	30-12-25	MONDAY	Learned in detail about Arduino, raspberry pi, esp8266
	31-12-25	TUESDAY	Learned in detail about different types og sensors like (temperature,humidity,motion) Actuators(relay,motor,led)
	1-1-25	WEDNESDAY	New year
	2-1-25	THURSDAY	Introduction to embedded C and python for IoT applications. Wrote and executed a simple program to blink n led using Arduino
	3-1-25	FRIDAY	Formation of project team Finalizing the idea of project(otp based smart locking system)

	DATE	DAY	NAME OF THE TOPIC / MODULE COMPLETED
3rd WEEK	6-1-25	MONDAY	Studied IoT applications in security system. Identified different components for project
	7-1-25	TUESDAY	Checked the compatibility of components
	8-1-25	WEDNESDAY	Designed initial circuit. Simulate circuit using proteus
	9-1-25	THURSDAY	Studied otp algorithms, decided to use randomly generated otp with gsm based sms delivery
	10-1-25	FRIDAY	Connected SIM800L GSM module to Arduino. Resolved network signal issues by adjusting power supply

	DATE	DAY	NAME OF THE TOPIC / MODULE COMPLETED
4th WEEK	13-1-25	MONDAY	Stored ge nerated otp in arduinos eeprom. Implemented keypad input for otp entry
	14-1-25	TUESDAY	Makarsankranti
	15-1-25	WEDNESDAY	Connected 16x2 LCD display with I2C module. Displayed equired messages
	16-1-25	THURSDAY	Completed all connections of hardware. Implemented and tested code
	17-1-25	FRIDAY	Presentation of project infront of project manager

5th WEEK	DATE	DAY	NAME OF THE TOPIC/MODULE COMPLETED
	20-1-25	MONDAY	Completed documentation process.took guidance from company people for future reference and a Had conversation with them regarding completed project.
	21-1-25	TUESDAY	Final day at company. Collected the internship completion certificate

1. Introduction

In today's technologically advanced world, security plays a crucial role in safeguarding our belongings. Traditional mechanical locks are no longer sufficient to meet the increasing demands for enhanced security and convenience. As a result, innovative solutions like the OTP (One-Time Password) based smart wireless locking system have emerged. The OTP based smart wireless locking system utilizes the power of Arduino, a popular open-source electronics platform, to create a secure and efficient locking mechanism. This system eliminates the need for physical keys and instead relies on a unique one-time password generated for each access attempt. By leveraging wireless communication technologies such as Bluetooth or Wi-Fi, the smart locking system enables users to control and monitor their locks remotely. It is connected to a reliable and secure OTP generation algorithm that generates a unique password for each access attempt, ensuring maximum security against unauthorized access. To operate the system, users simply need to enter the generated OTP through a smartphone application or a dedicated control panel. The Arduino verifies the OTP and triggers the locking or unlocking mechanism accordingly. Furthermore, the smart wireless locking system offers additional features such as real-time notifications, access logs, and the ability to grant temporary or time-limited access to authorized individuals.

2. Title/ Problem statement/ objectives

Title

Otp based smart locking system using arduino

Problem Statement:

In traditional locking systems, security is often limited to physical keys or password-based access, which can be easily compromised or duplicated. Furthermore, remote access or control of locks in many systems is either non-existent or lacks sufficient security measures. With the increasing demand for secure, remote, and easily manageable access control systems, there is a need for a more robust and secure solution. The challenge is to design a smart locking system that ensures user safety while enabling secure remote access, reducing the risk of unauthorized entry. This project addresses these challenges by leveraging the Internet of Things (IoT) and One-Time Passwords (OTP) to create a more secure, flexible, and efficient locking mechanism.

Objectives:

1. **Design and Develop an IoT-based OTP Locking System**:
   - Integrate Arduino with IoT components (e.g., GSM module, keypad etc) to create a functional and secure locking system.
2. **Generate and Validate OTPs Securely**:
   - Implement OTP generation and validation logic through GSM module to ensure that each access request is validated with a unique, time-sensitive password.
3. **Enable Remote Locking and Unlocking**:
   - Use SMS OTP delivery to enable remote authentication and control of the locking system, providing users with a convenient and secure method of access.
4. **Enhance Security Features**:
   - Ensure that the OTP system cannot be easily bypassed or compromised. Include features such as OTP expiration and attempts limitation to reduce the risk of unauthorized access.
5. **Ensure Real-time Functionality**:
   - Develop the system to respond in real-time to user inputs (i.e., entering OTPs on the keypad), with minimal delay between OTP generation and access authorization
   -

3. Motivation

The motivation behind developing an IoT-based OTP (One-Time Password) Locking System using Arduino stems from the growing need for secure, convenient, and modern access control systems. Traditional lock systems, relying on physical keys or static passwords, are becoming increasingly vulnerable to security breaches, such as unauthorized duplication or hacking.

Scope

In the age of IoT, the integration of smart systems into everyday applications presents an opportunity to enhance security and user experience. One-Time Password (OTP) systems, which generate a unique password for each login session, provide a high level of security by minimizing the risk of interception or unauthorized access. Combining OTPs with an IoT-based locking mechanism creates a secure, scalable, and easily manageable solution for remote and real-time access control.

4. Methodology details

The authentication technique used here could be an OTP (four-digit numeric) code generated in an Arduino microcontroller and sent to the registered mobile range through the GSM module and conjointly keep in the Arduino microcontroller's RAM, that is then entered through the computer keyboard.

The code entered this manner is then compared to the countersign keep in memory. The Arduino microcontroller endlessly monitors the computer keyboard for a match with the keep counter sign. As and once there's a match the output line is enabled which may then be wont to run the motor. Associate in Nursing liquid crystal {display| LCD digital show |alphanumeric display}The display is additionally wont to display whether or not the entered countersign is correct or not.

5. Results & conclusion

The system prompt the user to enter a valid OTP to gain access. The OTP is be generated through a mobile app. The Arduino board utilizes Bluetooth to receive the OTP from a smartphone The Arduino board will verify the received OTP against a pre-shared secret or a database of valid OTPs. If the OTP matches, the system will grant access by unlocking the door or activating an electronic lock mechanism. The system securely store the OTPs, ensuring that they are not easily accessible to unauthorized individuals. Techniques like encryption or hashing can be employed to enhance security. The system maintain a log of access attempts, including successful and unsuccessful ones. This log can be useful for monitoring and auditing purposes. This system include features to add, modify, or remove users and their corresponding OTPs. This is achieved through a dedicated mobile application. The overall success of the OTP-based smart wireless locking system will depend on the implementation quality, security measures, and reliability of the hardware and software components involved. It's important to thoroughly test and validate the system to ensure its effectiveness and robustness.

Many times we forgot to carry the key of our home. Or sometimes we come out of our home and door latch closes by mistake. In these cases it is really difficult to get inside the house . This project will help in keyless entry and at the same time will be more secure. This idea will minimize the overall cost by the use Bluetooth instead of GSM (which charges for providing service).Smart door lock is one of the most popular digital consumer devices due to its ease of use and affordable price. In fact, it replaces many common types of locks. This is project is good enough to provide security as long as password not shared and project is completely based on Android platform which is free open source software, So implementation rate is also inexpensive and easy to install anywhere. Its main advantage is to open door lock using an android where password is encrypted and home owner’s mobile phone will be notified every time the door opens. Hence, the project can be achieved in lesser time compared to other techniques previously employed.

6. Attendance Record

Learning Outcomes

{Some possible points to be covered in the title "Learning Outcomes" are:}

Understanding of IoT Integration:

- Gained hands-on experience in integrating various IoT components (Arduino, GSM module, relay, keypad) to create a secure, connected access control system.

Proficiency in Arduino Programming:

- Developed skills in programming the Arduino using **C** to handle multiple tasks such as reading user input from the keypad, generating OTPs, sending messages via GSM, and controlling hardware components.

Implementation of Secure OTP Systems:

- Learned the principles of **One-Time Password (OTP)** generation, including random number generation, time-sensitive expiration, and security best practices for preventing OTP reuse or interception.

Hands-on Experience with GSM Module:

- Developed a solid understanding of how to interface and program a **GSM module** for SMS-based communication. This includes sending and receiving OTPs to users' mobile phones, allowing for secure and remote access.

Working with Hardware Components:

- Gained practical knowledge of interfacing with external hardware components to build a complete access control system.

Project Management and Documentation:

- Learned how to document the design process, including system architecture, hardware setup, software development, and testing, to ensure clear communication and a structured approach to project development.

References

{Related to work}

{Some possible points to be covered in the title "References" are :}

[1] Pradnya R. Nehete, J. P. Chaudhari,S. R. Pachpande Door lock security systems

[2] Mr. Patil Bhushan, Mr. Mahajan Vishal, Mr. Pawar Mayur Automatic door lock system

[3] S. Umbarkar, G. Rajput, S. Halder, P. Harname and S. Mendgudle Automatic door lock system

[4] Aishwarya I P A survey on smart door lock security methodologies

[5] Snehalata Raut, Dimple chapke , Akash Sontakke , Nayan Pounikar , Prof. Neha Israni Door automation security system using OTP.

[6] Amanpreet Singh, Adarsh Sachan, Kashish Gupta, Gautam Kapoor, Harsh Kumar Singh, Ananya Singh IOT Based Smart lock